CHAPTER 2

THE WORLD POPULATION: DISTRIBUTION, DENSITY AND GROWTH

CHAPTER ESSENCE AT A GLANCE:
Examine the uneven spatial spread of global population, geographical and socio-economic determinants of density, mechanisms of demographic change, and the three stages of demographic transition theory.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Patterns of Global Population Distribution:** World population distribution is highly uneven; roughly 90 percent of humankind resides across barely 10 percent of total land area, heavily concentrated in Northern Hemisphere river basins.

• **Arithmetic Density of Population:** The arithmetic ratio of total population to total surface area provides a coarse measure of human spatial concentration across global regions.

• **Components of Population Change:** Population change is driven by crude birth rate (CBR), crude death rate (CDR), and migration. Growth can be natural (births minus deaths) or actual (factoring net migration).

• **Demographic Transition Theory:** Describes the historical shift from high birth and death rates in agrarian societies (Stage 1), through high fertility and falling mortality causing rapid growth (Stage 2), to low fertility and low mortality in urbanized industrial societies (Stage 3).

ESSENTIAL GEOGRAPHICAL MODELS & LANDFORMS

$$\text{Crude Birth Rate (CBR)} = \frac{\text{Live Births in a Year}}{\text{Mid-Year Total Population}} \times 1000$$

Crude Birth Rate per Thousand Population

$$\text{Crude Death Rate (CDR)} = \frac{\text{Total Deaths in a Year}}{\text{Mid-Year Total Population}} \times 1000$$

Crude Death Rate per Thousand Population

$$\text{Arithmetic Population Density} = \frac{\text{Total Population}}{\text{Total Surface Land Area}}$$

Arithmetic Population Density Ratio

$$\text{Natural Growth of Population} = \text{Crude Live Births} - \text{Crude Total Deaths}$$

Natural Growth Equation without Migration

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** Crude Birth Rate and Crude Death Rate are calculated per thousand persons of the mid-year population, not as percentage shares per hundred persons.

• **Alert:** Do not confuse Push factors with Pull factors; push factors make the place of origin unfavourable, whereas pull factors make the destination attractive.

• **Alert:** In Stage 1 of demographic transition, population growth is stationary not because births are low, but because very high mortality offsets very high fertility.

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Define Crude Birth Rate (CBR) and Crude Death Rate (CDR). [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Crude Birth Rate (CBR):** Expressed as the number of live births in a year per thousand of the mid-year population in a given region. **[1 Mark]**

• **Crude Death Rate (CDR):** Expressed as the number of deaths in a year per thousand of the mid-year population in a given region. **[1 Mark]**

2. Distinguish between natural growth of population and actual growth of population. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Natural Growth:** The difference between the number of live births and deaths in a specific territory over a given period (Natural Growth = Births - Deaths). **[1.5 Marks]**

• **Actual Growth:** The net demographic change incorporating natural growth and territorial migration (Actual Growth = Births - Deaths + In-migration - Out-migration). **[1.5 Marks]**

3. Differentiate between push factors and pull factors of migration with two examples of each. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Push Factors:** Conditions that compel individuals to abandon their home territory, such as acute unemployment, poor living conditions, political instability, and natural disasters. **[1.5 Marks]**

• **Pull Factors:** Attributes that attract migrants to a new destination, such as lucrative employment opportunities, higher wages, superior education, security of life, and pleasant climate. **[1.5 Marks]**

4. Explain the three stages of Demographic Transition Theory in detail. [Score: 4 Marks]

Answer & Key Valuation Steps:

• **Stage 1 (High Stationary):** Characterized by high fertility and high mortality; population growth is low and slow; predominantly rural agrarian society with low technological development. **[1.5 Marks]**

• **Stage 2 (Expanding):** Fertility remains high initially while mortality declines sharply due to improvements in sanitation, healthcare, and epidemic control, leading to high population growth. **[1.5 Marks]**

• **Stage 3 (Low Stationary):** Both fertility and mortality decline significantly; population becomes stable or grows slowly; society is urbanized, industrial, and literate with family planning. **[1 Mark]**

5. Describe any three geographical factors that influence the distribution of population in the world. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **1. Availability of Fresh Water:** Human settlements concentrate where fresh water is plentiful for drinking, irrigation, and navigation, such as river valleys. **[1 Mark]**

• **2. Landforms and Relief:** Flat plains and gentle slopes favour agriculture, transport, and factory construction, while rugged mountains deter human habitation. **[1 Mark]**

• **3. Climate and Soils:** Moderate temperate climates and fertile alluvial soils attract dense settlements, whereas extreme hot deserts and frozen tundras remain sparse. **[1 Mark]**

6. What is population doubling time? Explain how the time taken by world population to double has rapidly decreased over centuries. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Definition:** The duration in years required for a given human population to double its total demographic size at a constant annual growth rate. **[1 Mark]**

• **Historical Acceleration:** It took more than a million years for human population to reach one billion, but the increase from 5 billion to 6 billion took barely 12 years. **[1 Mark]**

• **Causal Catalysts:** Scientific inventions, modern hygiene, antibiotics, and mechanized food production drastically lowered global death rates, shortening doubling intervals. **[1 Mark]**

7. Why is arithmetic density of population an inadequate measure of the human-land relationship? [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Uniformity Fallacy:** Assumes human populations are uniformly distributed across all land area, overlooking vast uninhabitable zones like glaciated mountains and arid deserts. **[1 Mark]**

• **Carrying Capacity Deficit:** Fails to differentiate between productive arable agricultural land and uncultivable wasteland, hiding the actual ecological pressure on arable soils. **[1 Mark]**